

Publication

GAM: Generalized action model for robotic manipulation

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We present the Generalized Action Model (GAM), a production-grade foundation model that unifies robotic action generation across diverse tasks and embodiments through a vision-language-action (VLA) pipeline. GAM addresses two fundamental barriers in scaling robotic manipulation: the lack of a unified representation for diverse robot end-effectors and the prohibitive cost of acquiring high-quality interaction data at scale. Our approach introduces (1) a unified language-prompted policy and critic that generates and scores diverse manipulation actions—including suction grasps, pinch grasps, caging, and placements—from a single model, (2) a scalable offline data generation pipeline that recomputes dense action candidates and quality labels in simulation from real-world observations, and (3) an end-effector encoding that enables zero-shot transfer to unseen hardware. We validate GAM on a fleet of robotic workcells, where it has executed over 10 million pick-and-place cycles with greater than 95% pick and greater than 90% place success rates. The same model generalizes to hybrid end-effectors with distinct grasping modes at greater than 90% success.

Research areas

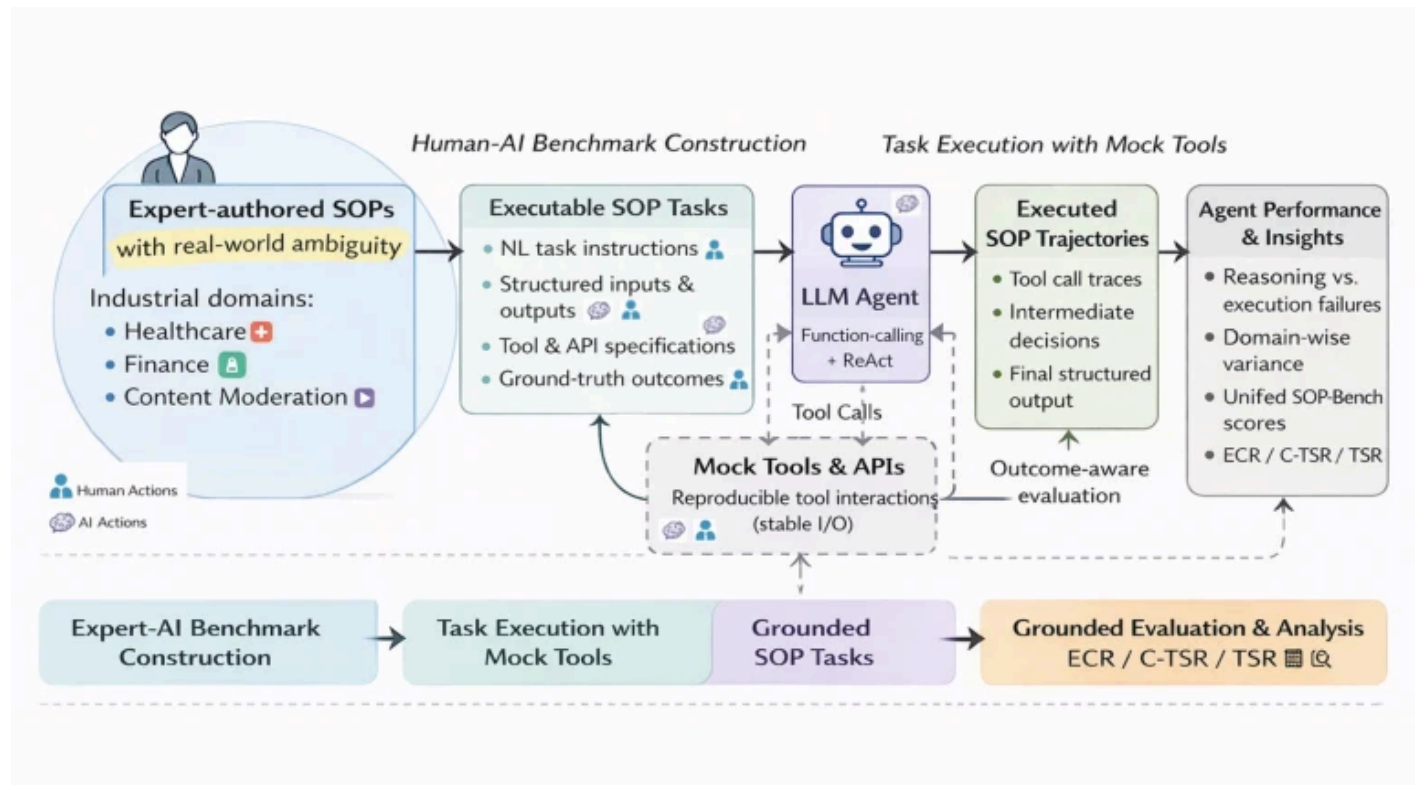
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